

SUHYEON YU

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EDUCATION

Rice University

Master of Computer Science (GPA: 3.91/4.00), Korean Graduate Student Association

Houston, TX, USA

Aug. 2025 – Dec. 2026

Sungkyunkwan University

B.S. in Computer Science and Engineering

Suwon, South Korea

Mar. 2018 – Aug. 2024

EXPERIENCE

Software Engineering Intern

Jun. 2026 – Aug. 2026

ABC Fitness

Austin, TX, USA

- **Frontend Modernization:** Rebuilt the user profile page from scratch in **React**, replacing jQuery as part of a company-wide modernization initiative for ABC Trainerize, which serves **400,000+ fitness professionals** and **1.6M+ clients** worldwide.
- **Async API Coordination:** Implemented asynchronous workflows with **RxJS** that coordinated multi-endpoint profile API requests, suppressed stale responses, and prevented overlapping operations across profile editing and uploading.
- **Unit Testing:** Wrote **Jest** unit tests to verify permission- and role-based access control, conditional rendering, and prevention of overlapping requests during photo/video uploads and profile updates, achieving **97% branch coverage** of profile business logic.
- **UI/UX Redesign:** Redesigned the user profile interface **beyond the original migration scope**, improving information hierarchy through a streamlined profile header, grouped content cards, and modal-based profile editing.
- **Cross-Functional Collaboration:** Collaborated with a UX designer to refine UI details and a QA analyst to verify functional correctness, delivering UX improvements such as drag-and-drop certification reordering to help trainers highlight their top credentials.
- **AI-Assisted Development:** Accelerated development by orchestrating multiple AI agents to build profile sections in parallel in separate Git worktrees and streamlined the development workflow using Figma **MCP** and team-shared agent skills.
- **Incremental Delivery:** Decomposed the profile rewrite into independently reviewable stacked PRs and used a **LaunchDarkly** feature flag for gradual rollout.

PROJECTS

Machine Unlearning Comparator | *React, TypeScript, D3, Zustand, Python, PyTorch*

- Built the frontend end-to-end for an interactive visualization dashboard for evaluating machine unlearning methods across accuracy, efficiency, and privacy using **React**, **TypeScript**, **D3**, and **Recharts**; project earned **100+ GitHub stars**.
- Designed the UI in **Figma** for a four-stage evaluation workflow spanning Build, Screen, Contrast, and Attack, around one A/B selection that drives class-wise accuracy, prediction matrices, layer-wise CKA, UMAP embeddings, and MIA simulation.
- Implemented 2,000-point penultimate-layer embedding scatterplots, held interaction latency to **$p95 \leq 11$ ms** via imperative D3 updates and memoization, and cut re-renders **~30%** by moving shared A/B state to **Zustand** selectors.

PyTorch-CKA | *Python, PyTorch*

- **AI Explainability:** Built an open-source Python library that compares internal representations across layers of **CNNs** and **LLMs**, addressing slow computation and memory buildup in existing CKA tools.
- **Performance Optimization:** Reduced HSIC trace complexity from **$O(B^3)$** to **$O(B^2)$** per layer pair (B: batch size) by replacing full matrix multiplication with Frobenius inner products, achieving an overall **$5.7\times$** speedup over an existing library on H100.
- **Memory Optimization:** Implemented automatic activation cleanup and forward-hook removal after CKA computation, reducing retained GPU memory by **95.7%** compared to an existing library (ResNet-18, batch size 1,000).
- **Deployment & Maintenance:** Published on PyPI (**~400 monthly downloads**) and automated CI testing across Python 3.10–3.14.

Online MyeongNyunDang | *Next.js, TypeScript, TanStack Query, Redux, Tailwind CSS, Docker, AWS*

- Built the frontend for an online education platform for students of my college using **Next.js**, **TypeScript**, **TanStack Query**, and **Redux**, which serves **5,000+** students, deployed on **Docker** and **AWS EC2**.
- Designed and implemented the admin and notification pages and migrated the styling system from Styled Components to **Tailwind CSS** to eliminate CSS-in-JS runtime overhead.
- Cut homepage initial load time by **~30%** with next/image lazy loading, caching with **TanStack Query**, and Next.js SSR/SSG.

PUBLICATION

J. Lee, S. Yu, Y. Jang, S. S. Woo, J. Jo. *Unlearning Comparator: A Visual Analytics System for Comparative Evaluation of Machine Unlearning Methods*. **IEEE Transactions on Visualization and Computer Graphics**, vol. 32, no. 3, Mar. 2026.

SKILLS & TOOLS

Languages: TypeScript, JavaScript, Python | **Frontend:** HTML, CSS, React, Next.js, Redux, TanStack Query, Tailwind CSS, SCSS, D3, Zustand
Backend: Node.js, FastAPI, GraphQL, REST APIs | **Databases:** SQL (PostgreSQL, MySQL), NoSQL (MongoDB) | **Testing:** Vitest, Jest, pytest
Cloud & DevOps: AWS (EC2, S3, Lambda), CI/CD (GitHub Actions), Docker, Git, GitHub | **AI Tools:** Cursor, Claude Code, Codex, Grok Build